v158TH FIGHTER WING
HOME OF THE GREEN MOUNTAIN BOYS
THE GREEN MOUNTAIN BOYS
HQ VERMONT AIR NATIONAL GUARD

STRIKE!

Beyond Visual Range

MAR 2026

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v158 FW

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- This lesson is **NON SENSITIVE**
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Beyond Visual Range (BVR)

Intercepts

Intercepts are a series of maneuvers, whether using a TAC/C2 node, GCI, AWACS, sensor fusion, or dead reckoning, that place the aircraft in a **position** from which a **weapon may be employed**. The type of **intercept geometry** utilized is based on various factors, including but thought not limited to pilot experience and **proficiency**, theatre **context**, available **sensors** or sensor fusions, and electronic warfare **(EW)**.

The intercept problem involves using sensor fusion to **detect** a target, then using geometry to arrive at a position from which the target can be **identified**, and **weapons fired** if necessary. At a minimum, the following tasks should be accomplished during an intercept:

- Positive Identification (PID).
- Positive closure rate.
- Weapons parameters consideration.
- Context consideration.

Baseline Intercept

The **Baseline Intercept** is a foundational technique upon which more advanced tactical gameplans are formulated. The Baseline Intercept consists of gaining an energy advantage over the adversary, as well as building an **optimal intercept course**, often nose-to-nose in theory, but offset in actuality.

Potential energy is a key topic within aviation. Energy, in the context of an aircraft's positioning and velocity, can be derived from a **higher altitude and speed** of the aircraft against Earth's force of gravity and drag. In essence, a pilot should command the aircraft to a higher altitude and airspeed to build an **energy advantage** over the adversary.

Optimal intercept course is a form of lead pursuit that creates a point where the two aircraft are set to collide. Theoretically, given that the pair are co-altitude and co-speed, basic trigonometry can be used to build an optimal intercept course, wherein one aircraft's Aspect Angle (AA) can be added to the other aircraft's AA to add up to 180 degrees.

BVR Timeline

The **BVR Timeline** is a **tactic** for keeping the **adversary** aircraft at a **sufficient distance** for the **employment** of MRAAMs or LRAAMs. The goal in a BVR engagement is to **neutralize** the adversary's **weapons** solution, and stray from the **risks** associated with **BFM** and **ACM**.

Central to the execution of this concept is an understanding of how both friendly and adversary **weapons** behave throughout an engagement. Effective management of the various factors within a BVR engagement, including closure rate, aspect angle, time, and range, allow pilots to **shape the fight**, forcing the adversary into a **defensive** or **reactive** posture, and preserving their own **survivability** in the fight.

The keys to a successful prosecution in a BVR engagement are vast, but rely heavily on **gameplanning**, **employment**, and **communication**. The BVR timeline helps with a few said keys to success, and consist of a set of straightforward definitions and associated ranges that will be detailed in the following slides.

MTR & Sorting

Minimum Targeting Range (MTR) is the specific, pre-briefed range, typically in NMs, that each flight member must employ on an adversarial group to ensure **group elimination**, or support otherwise planned intercept flows. MTR is a decision tool for flight leads to directively target, or “SORT” **wingmen** at ranges greater than MTR, or **flight leads** within MTR.

Sorting is a BVR tactic that can be used within a flight to effectively decide which flight member has awareness, PID, and is responsible for an individual or group of adversaries. However, sorting can quickly result in tangled communications, leading to a dangerous situation for the entire flight, and, as such, proper brevity and sorting techniques must be utilized.

The specific **sorting plan** will be discussed on a sortie-by-sortie basis. As a default, priority goes Lead, Left, Right, Low. Additionally, the flight leader may task the flight with sorting a group or aircraft within a specific group.

TR, DOR & Shot Calls

Transition Range (TR) is the minimum range at which an MRAAM or LRAAM must be launched in order to achieve an active state, and be out no later than Desired Out Range.

Desired Out Range (DOR) is the range from the closest group where a fighter's OUT maneuver will defeat any weapons employed by the adversary, while simultaneously preserving distance to RECOMMIT with sufficient range.

Shot Calls are typically a radio transmission over a pre-briefed frequency that declare, at a minimum, a missile or simulated weapons employment is occuring. In this context, brevity is important. A shot call without brevity looks like this:

- [Callsign], AAM launched, [Bearing], [Range], [Altitude], [Aspect], [PID].

An example optimal shot call with brevity looks like this:

- MAPLE11, FOX 3 wedge group BRAA 320 for 36, Angels 27, track south, hostile.

DR, MSR & MAR

Decision Range (DR) is the range from the closest group at which a fighter can execute a briefed defensive maneuver for a set period of time in an attempt to defeat an adversary's weapon solution, and then maneuver to abort the engagement. Doing this will kinematically defeat any adversary weapons solution shot at the fighter, and momentarily keep the fighter outside the adversary's maximum WEZ.

Minimum Shot Range (MSR) is the minimum range from the closest group at which an AAM must be shot by the fighter to achieve an active tracking state, and the fighter to be OUT no later than MAR.

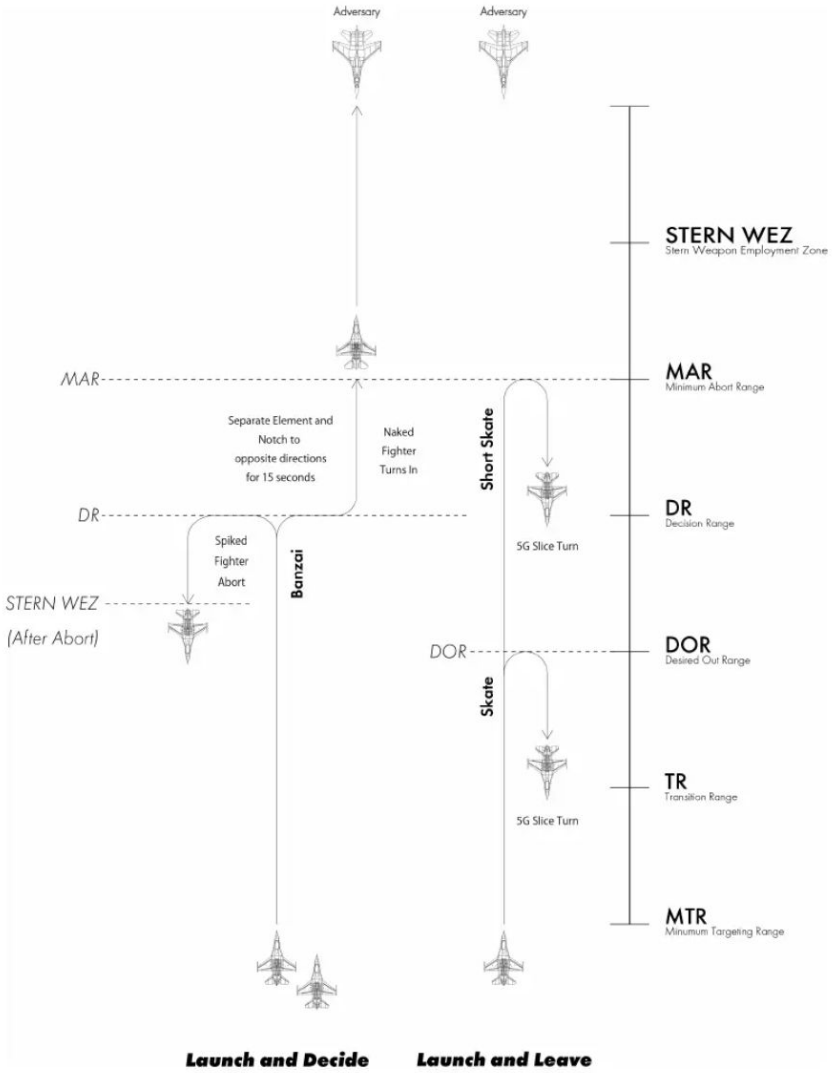
Minimum Abort Range (MAR) is the range from the closest group at which the fighter can abort from the engagement and remain outside of the adversary's WEZ.

MRR & STERN WEZ

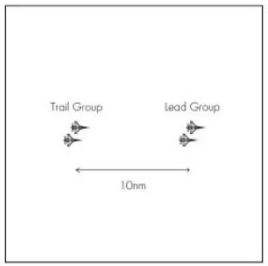
Minimum Recommit Range (MRR) is the minimum range from the closest group at which a fighter can RECOMMIT and turn IN, employ effective targeting, and make a shot with an AAM while remaining outside of STERN WEZ.

Stern Weapons Engagement Zone (STERN WEZ) is the maximum range at which an adversary's shot will hit a fighter from behind, assuming the fighter is COLD.

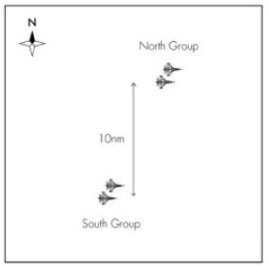
Timeline Diagram



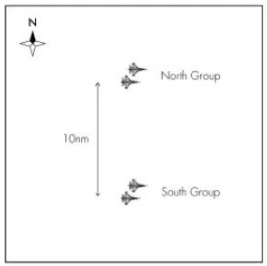
Group Types



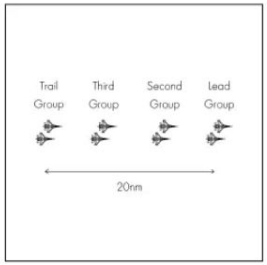
2 GROUP RANGE 10



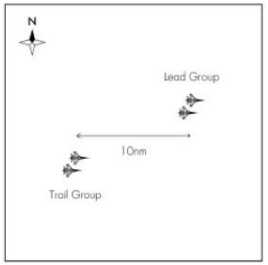
2 GROUP AZIMUTH 10
ECHELON SOUTH WEST



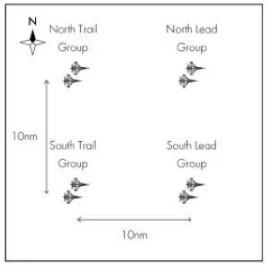
2 GROUP AZIMUTH10



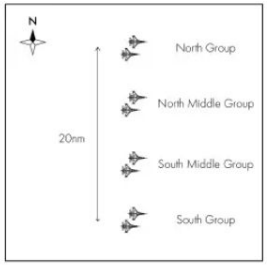
4 GROUP LADDER
20 DEEP



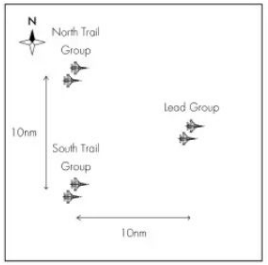
2 GROUP RANGE 10
ECHELON SOUTH WEST



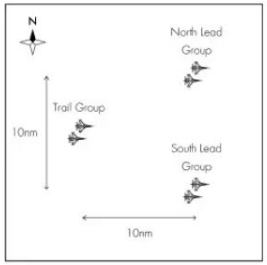
4 GROUP BOX
10 WIDE 10 DEEP



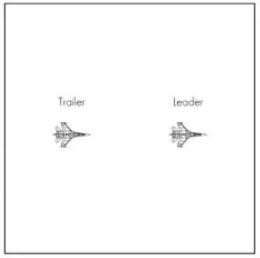
4 GROUP WALL
20 WIDE



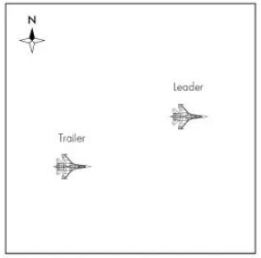
3 GROUP VIC
10 DEEP 10 WIDE



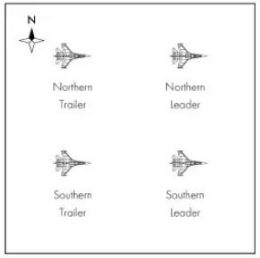
3 GROUP CHAMPAGNE
10 WIDE 10 DEEP



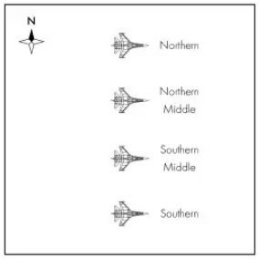
LEAD TRAIL



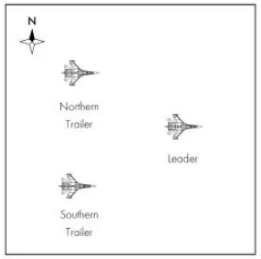
SWEPT



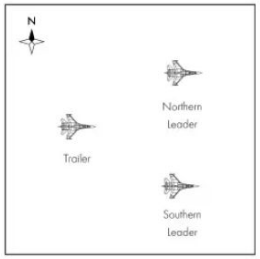
CONTAINER



LINE ABREAST



WEDGE



STINGER

Additional Study Material

- *Introduction to Beyond Visual Range Combat* by Mike Solyom
- *BVR Comms Explained* by Mike Solyom
- *Intercept* by Cider C

End

You have completed this lesson portion covering BVR.